

PAPER_297 UQFF Superluminal Hubble Expansion Ratio $\hat{I}_{exp} = 3.328 > 1$

First UQFF Module Where $v_{exp}/c > 1$ at System Boundary

Session: 84

Module: UNIVERSE_DIAMETER_UQFF_MODULE.cpp (26th C++ UQFF module "Observable Universe as System")

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Classification: Unique Physics "First UQFF Superluminal Expansion Parameter ($\hat{I}_{exp} > 1$)"

Abstract

The UQFF Observable Universe Diameter Module is the **first UQFF module** where the system boundary recession velocity exceeds the speed of light: $v_{exp} = H_0 \times r_{obs} = 9.984 \times 10^8 \text{ m/s} = 3.328c$. The dimensionless Superluminal Expansion Ratio $\hat{I}_{exp} = v_{exp}/c = 3.328 > 1$ is a new UQFF parameter encoding the cosmological property that the observable universe spans **3.328 Hubble lengths** ($r_{obs} = 3.328 \times r_H$). The Hubble-expansion coupling factor at t_{Hubble} is $(1 + H_0 \times t_H) = 1.988$ "a near-doubling of the Newtonian base over cosmic time. All 25 prior UQFF modules had $\hat{I}_{exp} < 1$ (sub-luminal expansion).

1. Physical Setup

System: Observable Universe

Observable universe radius: $r_{obs} = 4.4 \times 10^{26} \text{ m}$

Hubble constant: $H_0 = 70 \text{ km/s/Mpc} = 2.269 \times 10^{-18} \text{ s}^{-1}$

Speed of light: $c = 3.0 \times 10^8 \text{ m/s}$

Hubble sphere (Hubble radius): $r_H = c/H_0 = 1.322 \times 10^{26} \text{ m} = 4.28 \text{ Gly}$

Cosmic age: $t_H = 4.355 \times 10^{17} \text{ s} = 13.8 \text{ Gyr}$

2. Master Relation

Hubble recession velocity at observable universe boundary:

$$v_{exp} = H_0 \times r_{obs} = 2.269 \times 10^{-18} \times 4.4 \times 10^{26} = 9.984 \times 10^8 \text{ m/s}$$

Superluminal Hubble Expansion Ratio:

$$\eta_{exp} = \frac{v_{exp}}{c} = \frac{9.984 \times 10^8}{3.0 \times 10^8} = 3.328 > 1$$

Hubble length ratio:

$$\frac{r_{obs}}{r_H} = \frac{v_{exp}}{c} = \eta_{exp} = 3.328$$

The observable universe boundary is 3.328 Hubble lengths from Earth – comfortably beyond the Hubble sphere where recession velocity equals c .

3. Hubble-UQFF Near-Doubling of Gravity

The base gravity term with Hubble expansion coupling:

$$a_{base}(t) = g_{base} \times (1 + H(z) \times t) = 3.447 \times 10^{-10} \times (1 + 2.269 \times 10^{-18} \times t)$$

At $t = t_H = 4.355 \times 10^8$ s:

$$a_{base}(t_H) = 3.447 \times 10^{-10} \times (1 + 0.988) = 3.447 \times 10^{-10} \times 1.988 = 6.854 \times 10^{-10} \text{ m/s}^2$$

Hubble expansion factor $\hat{I}_{\frac{3}{4}H} = 1 + H \times t_H = 1.988$:

This near-doubling factor (≈ 2) reflects that the UQFF base gravity **almost doubles** over the Hubble time when the Hubble coupling is included – a striking result confirming that the Universe-scale Hubble coupling is an $O(1)$ effect (not a small correction).

4. Superluminal Expansion in the EM Lorentz Term

The EM Lorentz term explicitly incorporates $\hat{I}_{exp}$:

$$a_{EM} = \frac{q \cdot v_{exp} \cdot B_{cosmic}}{m_p} \times (1 + \eta_{exp}) \times \text{scale}_{EM}$$

With $\hat{I}_{exp} = 3.328$, the factor $(1 + \hat{I}_{exp}) = 4.328$ vs. $(1 + 1) = 2$ for a subluminal system. This introduces a **42% EM enhancement** relative to a hypothetical c -speed boundary reference.

Numerically:

$$\begin{aligned} a_{EM} &= \frac{1.602 \times 10^{-19} \times 9.984 \times 10^8 \times 10^{-15}}{1.673 \times 10^{-27}} \times 4.328 \times 10^{-12} \\ &= 95.59 \times 4.328 \times 10^{-12} = 4.136 \times 10^{-10} \text{ m/s}^2 \end{aligned}$$

The EM term ($4.136 \times 10^{-10} \text{ m/s}^2$) is **comparable to the Newtonian base** ($3.447 \times 10^{-10} \text{ m/s}^2$) – another first for UQFF modules.

5. Special Relativity Compatibility

The superluminal recession velocity $v_{\text{exp}} > c$ does NOT violate special relativity. This is a **coordinate velocity** (metric expansion), not a proper velocity between local inertial frames. In GR, the expansion of the universe allows coordinate distances to grow faster than c , as confirmed by the cosmological horizon structure. The Hubble-flow velocity $\hat{I}_{\text{exp}} > 1$ is part of the cosmological metric, not a violation of local Lorentz invariance.

Specifically, this corresponds to objects beyond the **Hubble sphere** ($r > r_H$) in co-moving coordinates. For the observable universe at $r = 4.4 \times 10^{26}$ m with $r_H = 1.322 \times 10^{26}$ m, we are 3.33 Hubble lengths out $\hat{\square}$ solidly in the superluminal expansion regime.

6. $\hat{I}_{\text{exp}}$ Parameter in UQFF Architecture

Module	Session	r_{obs} (m)	v_{exp}/c ($\hat{I}_{\text{exp}}$)	$\hat{I}_{\text{exp}} > 1$
SGR1745	65	~ 0.01 pc	~ 0 (local)	No
Pillars of Creation	68	3.3×10^{16} m	$\hat{\square} 1$	No
NGC1792	73	7.6×10^{20} m	$\hat{\square} 1$	No
Andromeda	75	1.04×10^{22} m	$\hat{\square} 1$	No
HUDF (z=3.5)	72g	1.23×10^{22} m	~ 9 (co-moving)	Yes (but not UQFF param)
Universe Diameter	84	4.4×10^{26} m	3.328	Yes $\hat{\square}$ FIRST explicit

The key distinction: prior modules used $H(z)$ as a correction factor, never explicitly computing or naming $\hat{I}_{\text{exp}}$ as a parameter. PAPER_297 establishes $\hat{I}_{\text{exp}}$ as a first-class UQFF parameter.

7. Hubble Sphere as UQFF Speed-of-Light Boundary

Define the **UQFF Hubble Horizon** as the radius where $\hat{I}_{\text{exp}} = 1$:

$$r_H = \frac{c}{H_0} = \frac{3 \times 10^8}{2.269 \times 10^{-18}} = 1.322 \times 10^{26} \text{ m} = 4.28 \text{ Gly}$$

Objects beyond r_H recede superluminally. The observable universe ($r_{\text{obs}} = 4.4 \times 10^{26}$ m) extends to 3.328 r_H $\hat{\square}$ confirming we can observe objects that are currently receding faster than light (their photons from early epochs can still reach us).

This adds a new entry to the UQFF speed-of-light boundary catalog: - **PAPER_264**: Anti-gravity boundary at $f_{\text{TRZ}} = -1$ (HUDF module) - **PAPER_266**: Meissner quench at $B = B_{\text{crit}}$ - **PAPER_297**: Hubble superluminal horizon at $r = r_H$ (**this paper**)

8. WOLFRAM Term

```
eta_exp=v_exp/c=H0*r/c=9.984e8/3e8=3.328>1;
FIRST UQFF eta_exp>1;
r_H=c/H0=1.322e26m Hubble sphere;
r_obs=3.328*r_H;
expansion_factor(t_H)=1+H*t=1.988(near-doubling);
EM term a_EM prop eta_exp [PAPER_297]
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9. Key Values Summary

Quantity	Symbol	Value	Unit
Hubble recession velocity	v_{exp}	9.984	m/s
Superluminal ratio	$\hat{I}_{\text{exp}}$	3.328 > 1	dimensionless
Hubble radius	r_H	1.322	m
Observable / Hubble ratio	r_{obs}/r_H	3.328	dimensionless
Hubble expansion factor	$\hat{I}^{3/4}_H$	1.988	dimensionless
EM term	a_{EM}	4.136	m/s ²

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UQFF computed: Canonical UQFF buoyancy parameter $U_{\text{bi}} = \gamma \times [\text{SSq}] \times \text{GM}/r^2 = 5.0\text{e-}4 \times 0.57 \times 6.67\text{e-}11 \times M/r^2$; for solar parameters: $U_{\text{bi,Sun}} = 5.7\text{e-}4 \times 6.67\text{e-}11 \times 1.99\text{e}30 / (6.96\text{e}8)^2 = 1.47\text{e+}2 \text{ m/s}^2$.